

RPC muography comparison - plots for approval

Kostadin Raychev

kostadin.raychev@cern.ch

INRNE-Sofia, Bulgarian Academy of Sciences

GENERAL MUON MEETING

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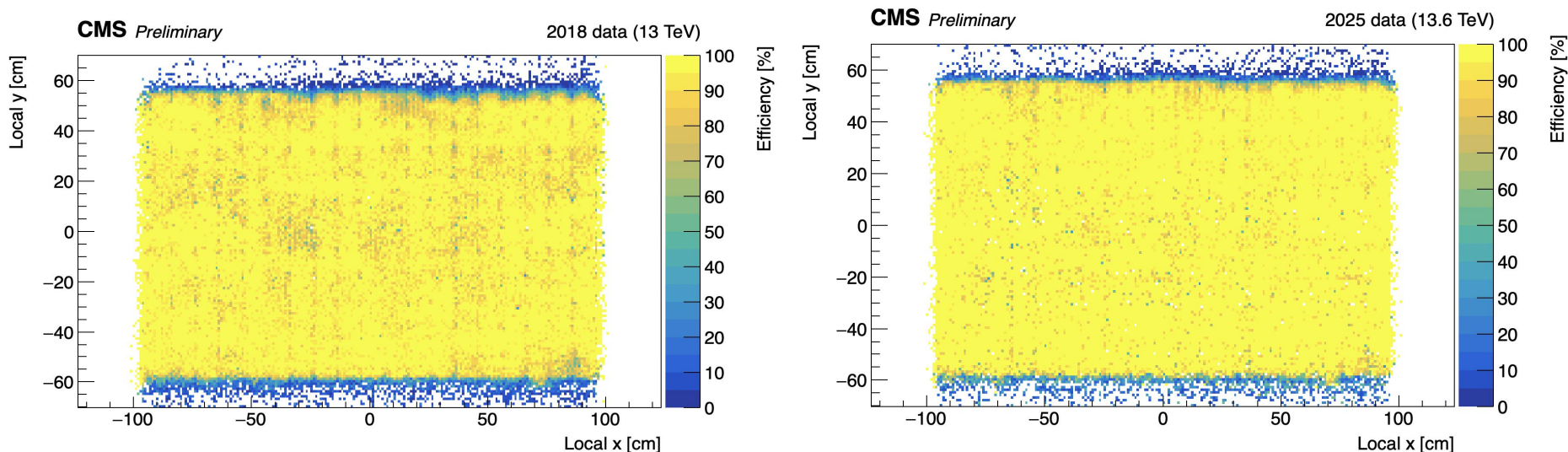
Introduction

- The RPC efficiency maps, so called muographies, have been obtained, exploring the Segment extrapolation method (JINST 5 (2010) T03017, DOI: 10.1088/1748-0221/5/03/T03017).
- The granularity of the muography plots is about $1 \times 1 \text{ cm}^2$.
- Two sets of proton-proton collision data from 2018 (Run-2) and 2025 (Run-3), taken at similar conditions, i.e. inst. luminosity $\sim 2.5 \text{E}34 \text{ cm}^{-2}\text{s}^{-1}$ and number of colliding bunches > 2000 have been used for the analysis.
- The muographies have been obtained for all the RPC rolls and the efficiency maps have been compared in order to study for any sign of potential ageing.
- A special attention is given to the RPC chambers, that have been switched Off because of the leaks, and later on have been repaired and switched back on.
- Apart from the visual comparison a simple statistical analysis comparing 2018 and 2025 data has been performed as well.
- For every 2D bin ($\sim 1 \times 1 \text{ cm}^2$) a relative efficiency difference $\Delta \epsilon_{ij}$ has been calculated, exploring the following formula:

$$\Delta \epsilon_{ij} = \{\epsilon_{ij}(2018) - \epsilon_{ij}(2025)\} / \{\epsilon_{ij}(2018) + \epsilon_{ij}(2025)\}$$

RPC efficiency vs local impact point

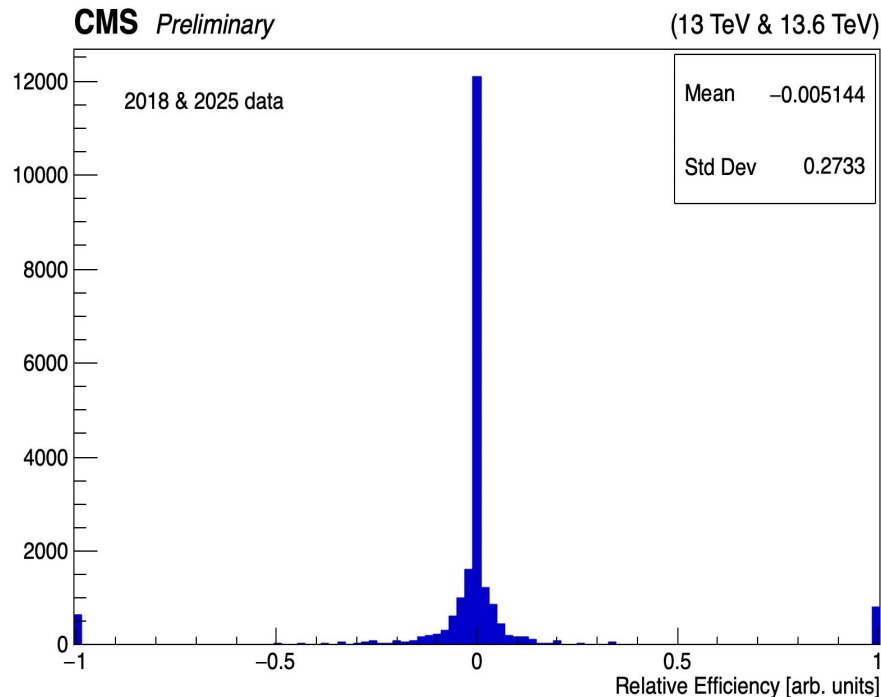
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The plots display the efficiency versus the local impact point on the RPC surface of a roll in one of the Barrel wheels (W-2_RB1in_S09_Backward), using data collected in 2018 (22.15 fb⁻¹ at 13 TeV - left plot) and 2025 (23.94 fb⁻¹ at 13.6 TeV - right plot). The lower efficiency spots are due to the dead regions induced by spacers (the two resistive electrodes are separated by 2 mm spacers to form the parallel plate gas gaps) on grid with dimensions with 10 × 10 cm². The Y axes is along the strip length. Due to the geometrical issues there are no extrapolated hits in the area corresponding to $x \geq 98$ cm, so the efficiency cannot be calculated for this area. Efficiency values were determined through the Segment Extrapolation method (JINST 5 (2010) T03017, DOI: 10.1088/1748-0221/5/03/T03017). This chamber operated continuously throughout Run 2. The plot on the left was taken at the end of Run 2 of the Large Hadron Collider (LHC). This chamber was then disconnected from the gas supply due to a gas leak in the corresponding gas channel (two chambers are connected per gas channel). It was reconnected to the gas supply in 2023 and the gas input pipes were changed during the Year-End Technical Stop (YETS) 2023/2024. The plot on the right was made after the gas pipes replacement procedure and more than one year of operation. This plot shows that the chamber has stable performance once it receives an appropriate gas flow. Contact: cms-dpg-conveners-rpc@cern.NOSPAMPLEASE.ch

Relative efficiency difference distribution

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Relative efficiency difference distribution. A bin-by-bin comparison of the RPC efficiencies measured in 2018 and 2025 is performed using the normalized difference $(\text{Eff}_{2018} - \text{Eff}_{2025}) / (\text{Eff}_{2018} + \text{Eff}_{2025})$. This quantity is bounded between -1 and $+1$, where a value of 0 indicates identical efficiencies in the two data-taking periods. A value of $+1$ corresponds to bins where the 2025 efficiency is zero while the 2018 efficiency is non-zero, whereas -1 corresponds to bins where the 2018 efficiency is zero while the 2025 efficiency is non-zero. As a conclusion, the current results demonstrate that the chamber has shown a stable and high efficiency performance after being put back into operation following the gas repair procedure.